CAMBRIDGE INTERNATIONAL EXAMINATIONS

International General Certificate of Secondary Education

MARK SCHEME for the May/June 2014 series

0625 PHYSICS

0625/22 Paper 2 (Core Theory), maximum raw mark 80

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge will not enter into discussions about these mark schemes.

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NOTES ABOUT MARK SCHEME SYMBOLS & OTHER MATTERS

B marks are independent marks, which do not depend on any other marks. For a B mark to be scored, the point to which it refers must actually be seen in the candidate's answer.

M marks are method marks upon which accuracy marks (A marks) later depend. For an M mark to be scored, the point to which it refers **must** be seen in a candidate's answer. If a candidate fails to score a particular M mark, then none of the dependent A marks can be scored.

C marks are compensatory method marks which can be scored even if the points to which they refer are not written down by the candidate, provided subsequent working gives evidence that they must have known it, e.g. if an equation carries a C mark and the candidate does not write down the actual equation but does correct working which shows he knew the equation, then the C mark is scored.

A marks are accuracy or answer marks which either depend on an M mark, or which are one of the ways which allow a C mark to be scored.

c.a.o. means "correct answer only".

e.c.f. means "error carried forward". This indicates that if a candidate has made an earlier mistake and has carried his incorrect value forward to subsequent stages of working, he may be given marks indicated by e.c.f. provided his subsequent working is correct, bearing in mind his earlier mistake. This prevents a candidate being penalised more than once for a particular mistake, but **only** applies to marks annotated "e.c.f."

e.e.o.o. means "each error or omission".

Brackets ( ) around words or units in the mark scheme are intended to indicate wording used to clarify the mark scheme, but the marks do not depend on seeing the words or units in brackets, e.g. 10 (J) means that the mark is scored for 10, regardless of the unit given.

<u>Underlining</u> indicates that this <u>must</u> be seen in the answer offered, or something very similar.

OR/or indicates alternative answers, any one of which is satisfactory for scoring the marks.

Spelling Be generous about spelling and use of English. If an answer can be understood to mean what we want, give credit.

Significant figures
Answers are acceptable to any number of significant figures ≥ 2, except if specified otherwise, or if only 1 sig. fig. is appropriate.

Units Incorrect units are not penalised, except where specified. More commonly, marks are allocated for specific units.

Fractions These are only acceptable where specified.

Extras Ignore extras in answers if they are irrelevant; if they contradict an otherwise correct response or are forbidden by mark scheme, use right + wrong = 0.

Ignore indicates that something which is not correct is disregarded and does not cause a right plus wrong penalty.

Not/NOT indicates that an incorrect answer is not to be disregarded, but cancels another otherwise correct alternative offered by the candidate, i.e. right plus wrong penalty applies.

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1	**(a)**	area under graph OR ½(u + v) t	C1
		½ × 40 × 8	C1
		160 (m)	A1
	(b)	315 + candidate's **(a)**	C1
		distance = speed × time **OR** distance / time in words, symbols or numbers	C1
		(315 + 160) / 80 **OR** (315 + candidate's **(a)**) / 80	C1
		(5.9) 38(m/s)	A1
	(c) **(i)**	steady / same / constant / uniform speed	B1
	(ii)	slowing down / decelerating / negative acceleration	B1
			[Total: 9]
2	**(a)**	measuring cylinder / graduated cylinder	B1
	(b)	balance, accept spring balance, accept (weighing) scales	B1
	(c)	find mass of empty cylinder	B1
		find mass of cylinder + liquid	B1
		subtract values **NOT** if stated the wrong way round accept valid alternative methods	B1
	(d)	density = mass / volume, in words, symbols or numbers	C1
		62.4 ÷ 80	C1
		0.78 **OR** 780	A1
		g / cm^3 **OR** kg / m^3 as appropriate	B1
			[Total: 9]

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3	(a)	equal (size / magnitude) / the same (size), ignore opposite **NOT** same direction	B1
	(b)	it would (start to) sink (if weight>upthrust)	B1
	(c)	moves (forward)	C1
		accelerates forward / increases speed / moves faster	A1
	(d)	slows down, IGNORE stops (moving)	B1

[Total: 5]

4	(a)	idea of expansion / gets bigger	B1
	(b)	particles have more energy / vibrate faster ignore move quickly or move faster	B1
		particles move apart / space between particles increases **NOT** particles expand	B1
	(c)	contracts / gets smaller / shrinks IGNORE fits tightly	B1
	(d)	idea of being pushed together accept move / stick together / compressed accept pulled tight/together	B1

[Total: 5]

5	(a)	(i)	wax melts (faster) on copper rod	B1
			wax melts less (far) / not at all / slower on plastic rod comparison needed	B1
		(ii)	IGNORE any statements about <u>conduction of electricity</u> copper is a (good) (thermal) <u>conductor</u>	B1
			plastic is an <u>insulator</u> / poor <u>conductor</u>	B1
	(b)		(only) faster / high (k.)e. / most energetic particles	B1
			escape / go into the air or leave the water (surface)	B1
			(this means) average (k.) e. of water decreases/falls accept internal energy / thermal energy for k.e.	B1

[Total: 7]

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6	**(a)**	speed = distance / time in words, symbols or numbers **OR** distance / speed	C1
		330 / 5000	C1
		0.066 (s), allow 0.07 (s) (to one significant figure)	A1
	(b)	man with hammer hears one sound accept hears the sound almost instantly / first	B1
		other man hears two sounds **OR** 1 through air and 1 through rail **NOT** hears two sounds, one is an echo	B1
		any one from: hears sound through rail before sound through air calculation of time difference between sounds because (speed of sound) in metal/steel faster than (speed of) sound in air	B1
			[Total: 6]
7	**(a) (i)**	correct idea ± 1 line	C1
		correct distance	A1
	(ii)	(slinky spring) moved backwards and forwards owtte	B1
	(b) (i)	correct idea e.g. crest to crest **NOT** just 2 peaks marked	C1
	(ii)	idea of bigger (vertical) distance between crest and trough	B1
	(c) (i)	no change / nothing	B1
	(ii)	less / shorter / smaller / decreases	B1
			[Total: 7]
8	**(a) (i)**	any one from: aluminium, copper, gold, iron	B1
	(ii)	any one from: ebonite, glass, plastic, silk	B1
	(iii)	iron	B1
	(iv)	any one from: ebonite, glass, plastic, silk	B1

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(b) accept correct alternative methods

stroke with pole of magnet — B1

in one direction — B1

OR (alternative answer)
place in solenoid / coil — (B1)

current in one direction / battery / d.c. — (B1)

[Total: 6]

9 (a) (i) ammeter **NOT** ampmeter
accept multimeter on current range — B1

(ii) 2nd box ticked, current — B1

(b) (i) 1st box ticked, charge — B1

(ii) **1.** ($R =$) $R_1 + R_2$ in words, symbols or numbers — C1

24 (Ω) — A1

2. $V = IR$ in any form **OR** V/R — C1

12 / 24 e.c.f. — C1

0.5 e.c.f. — A1

A **OR** amp(s) **OR** ampere(s) — B1

(c) bottom box ticked, 0 V — B1

[Total: 10]

10 (a) lamp will blow / burn out
accept blow up / glow too / very brightly ignore bright / won't work — B1

(b) (i) transformer shown with one coil across input and other coil across output
accept any reasonable attempt at transformer symbol — B1

(ii) factor of 2 e.g. 12 / 6, 6 / 12 or 2:1 ignore units — C1

1:2 **OR** 1 to 2 — A1

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(c) **(i)** resistor shown joining top two wires or bottom two wires M1
accept diagonal connection

complete series circuit A1
note: 2 resistors in series gains only one mark B1

(ii) 1.5 (Ω) B1

[Total: 7]

11 **(a)** 23 B1

(b) 11 B1

(c) 12 B1

(d) 11 no e.c.f. from **(b)** B1

[Total: 4]

12 **(a)** 4 (hours) B1

appropriate indication of method (minimum indication any halving of count rate on axis or curve) B1

(b) **(i)** 1000 B1

(ii) candidate's **(a)** B1

(iii) in the range 62 – 63, e.c.f. from **(b)** **(i)** and **(b)(ii)** B1

[Total: 5]

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